

# ER SG04

Integrated Cases with Histology Cell Biology & Gross and Developmental  
Anatomy

## **Section A:**

### **Research Article – Self-directed learning – Urethral tears**

Use library resources or open resources (eg. Google Scholar, PubMed) to find a peer-reviewed article on Urethral tears. Use the following objectives as a guide to review the selected paper. Once you can answer most of these questions, then you have completed your review and are fully prepared for discussion.

- 1. Discuss the boundaries and contents of the superficial and deep perineal pouches**
- 2. Differentiate between anterior and posterior urethral tears**
- 3. Differentiate between extravasation of urine between anterior and posterior urethral tears**

# Section B: Integrated Cases

Histology Cell Biology & Gross and Developmental Anatomy

# Objectives

|                               |                                                                                                             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|
| SOM.MK.I.BPM2.2.ER.1.HCB.1109 | Identify and describe the cells, tissues, organs and associated structures of the Male reproductive system. |
| SOM.MK.I.BPM2.2.ER.1.HCB.1110 | Explain the key roles of the cells and organs associated with the Male reproductive system to the group.    |
| SOM.MK.I.BPM2.2.ER.1.HCB.1111 | Apply the histology of the Male reproductive system to clinical cases and correlations.                     |
| SOM.P.I.BPM2.2.ER.6.HCB.1112  | Develop both professional and presentation skills.                                                          |

# Integrated Case 1

A 48-year-old man comes to the physician because of a 6-month history of a progressively increasing swelling of the left scrotum. Physical examination shows an increase in the size of the left testes. Cord like structures which feels like a "bag of worms" are palpated in the left scrotal area when the patient is standing and asked to perform Valsalva's maneuver (seen in the attached image). Blood and semen are collected for analysis and scrotal ultrasound with Doppler are performed.



## Semen Analysis:

| Parameter   | Patient results       | Reference range         |
|-------------|-----------------------|-------------------------|
| Sperm count | 10,000000/ml          | = or > 20million/ml     |
| Viability   | 38% alive             | 75% or more             |
| Motility    | 32%                   | 50% or >                |
| Morphology  | 4% by Kruger criteria | >14% by Kruger criteria |
| WBC         | 500000/ml             | ≤ 1,000,000/ml          |
| Fructose    | 20uM/ejaculate        | 13uM/ejaculate          |
| pH          | 7.7                   | 7.2 - 8                 |

## Blood Results:

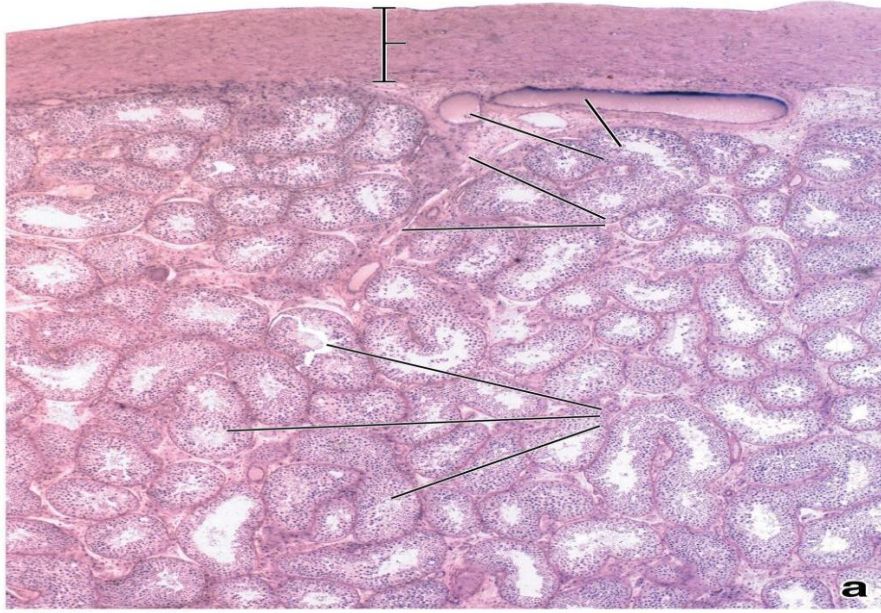
| Parameter    | Patient results | Reference range                   |
|--------------|-----------------|-----------------------------------|
| Testosterone | 200ng/dl        | 270 – 1200ng/dl (varies with age) |
| DHT          | 20ng/dl         | 27- 75ng/dl varies with age.)     |
| FSH          | 20mIU/ml        | 2-15U/L                           |
| LH           | 18mIU/ml        | 2-15U/L                           |

# Questions: Integrated Case 1

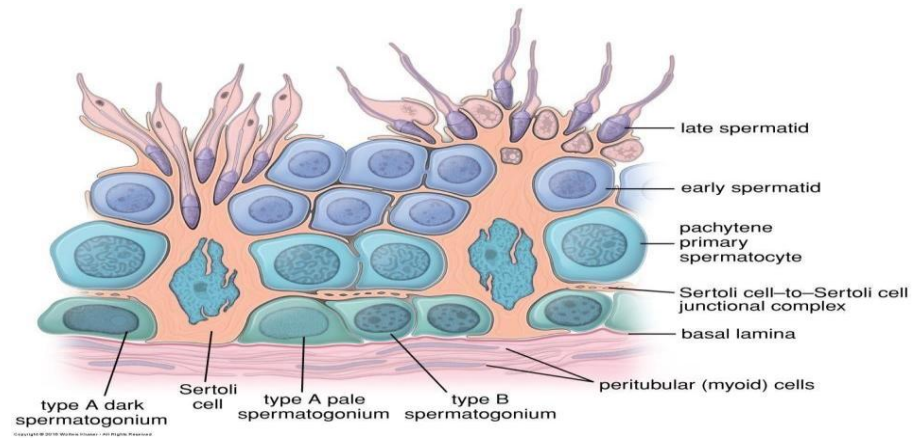
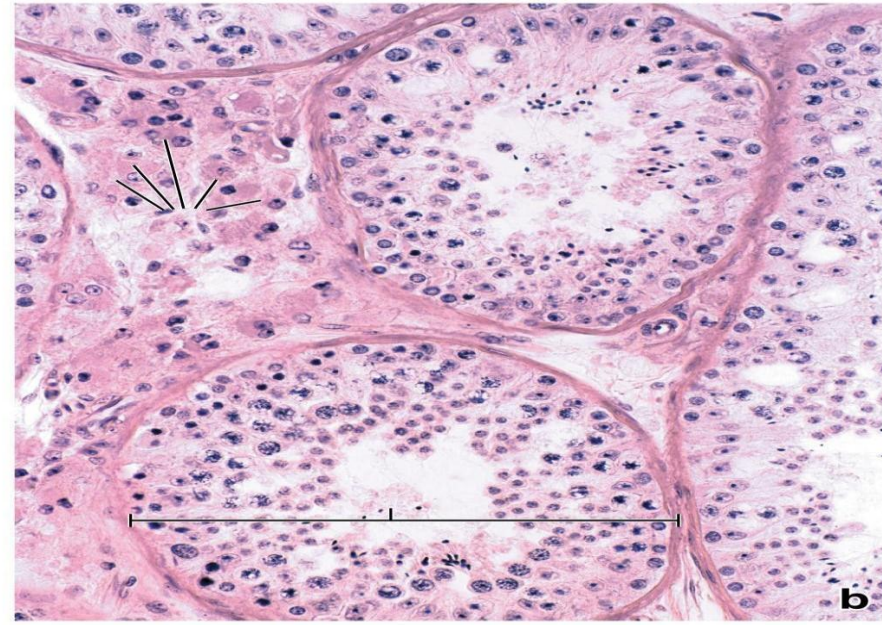
1. Analyze the lab results.
2. (a) Explain why the pituitary hormones are elevated.  
(b) What is the site of secretion of testosterone?
3. Using the following slides, list and describe the microscopic appearance of the organs/glands that contribute to the production of semen? Discuss their contributions. Include a review of the accessory glands that contribute to the production of semen.



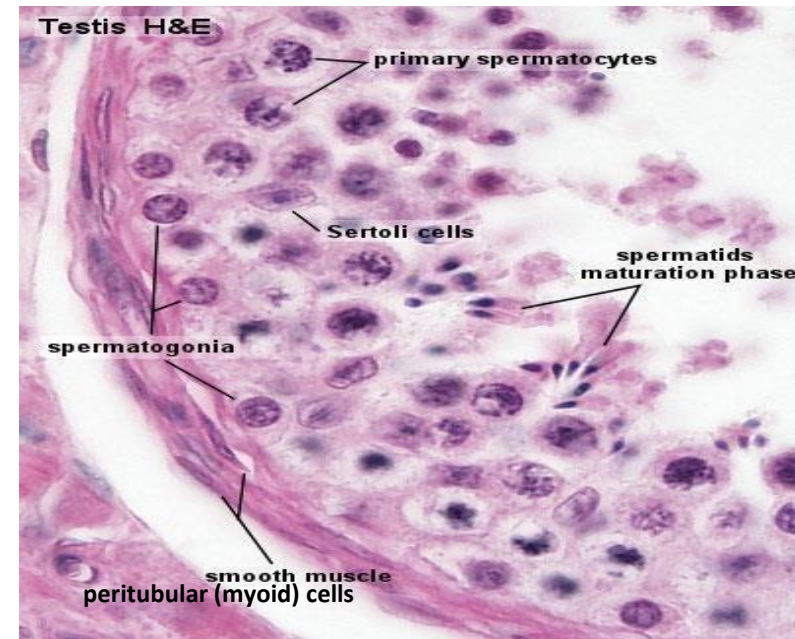
# Testis



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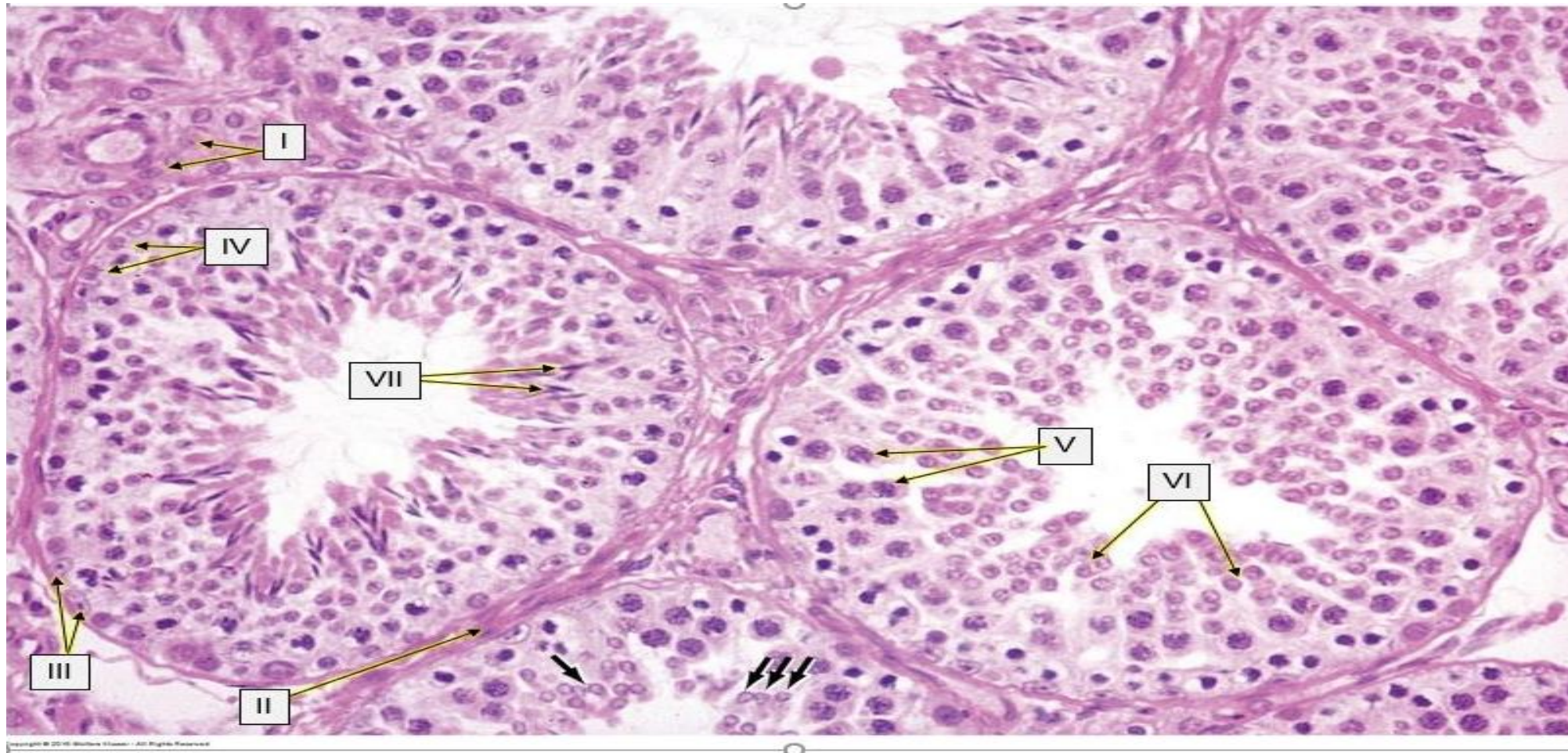


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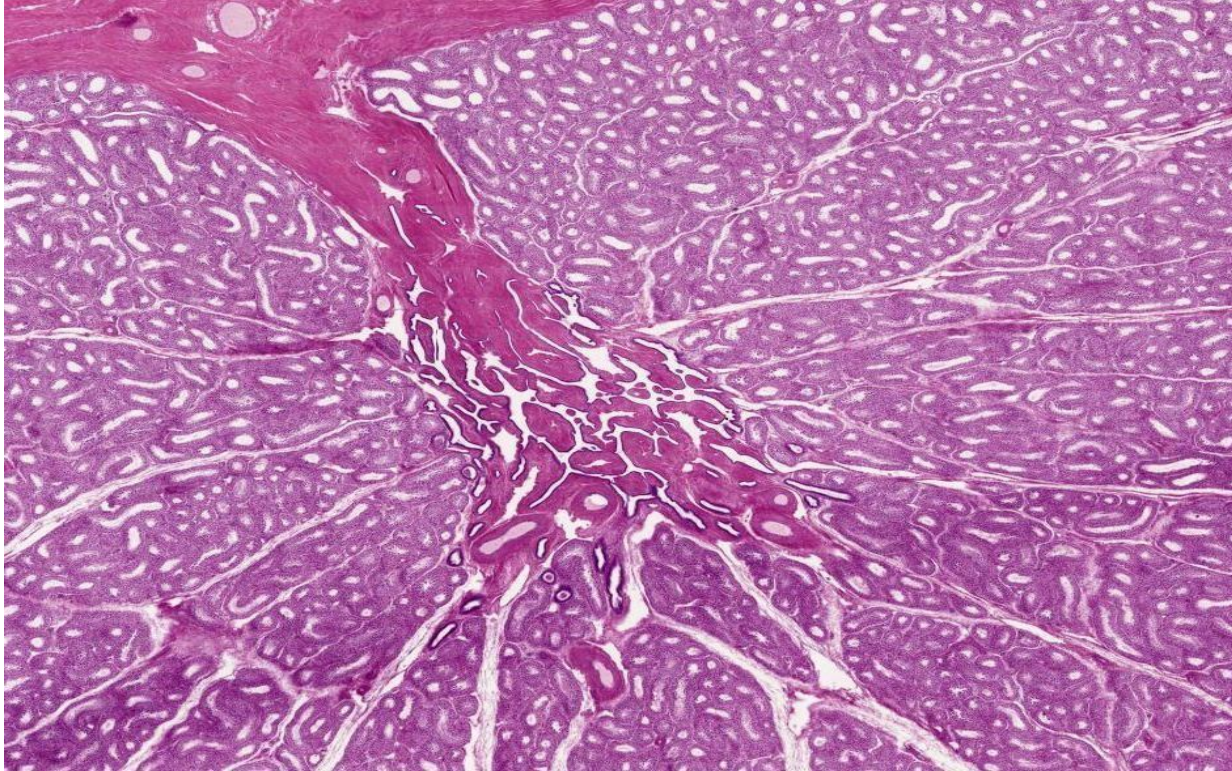


Identify the Roman Numerals (I – VII) below:

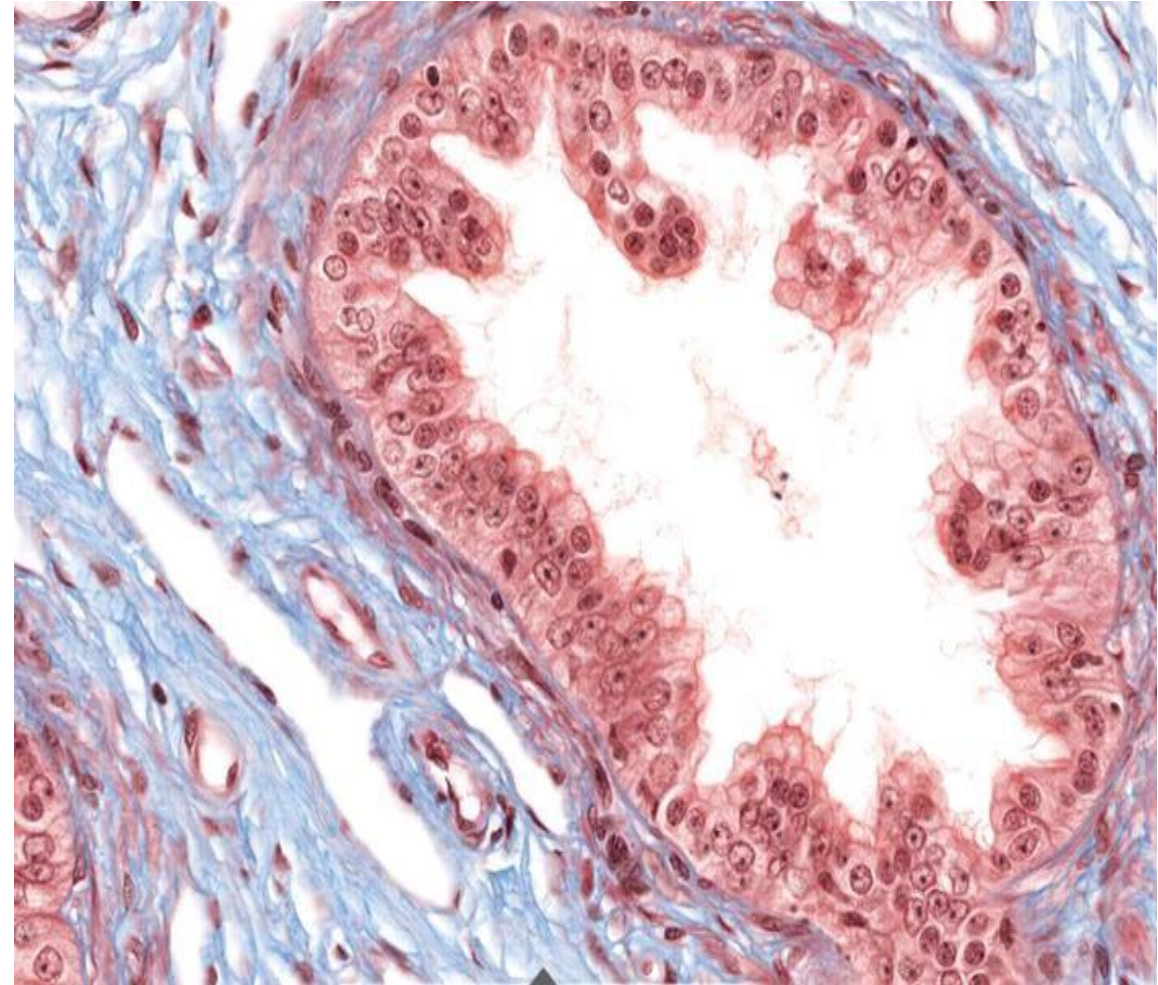




Rete testis

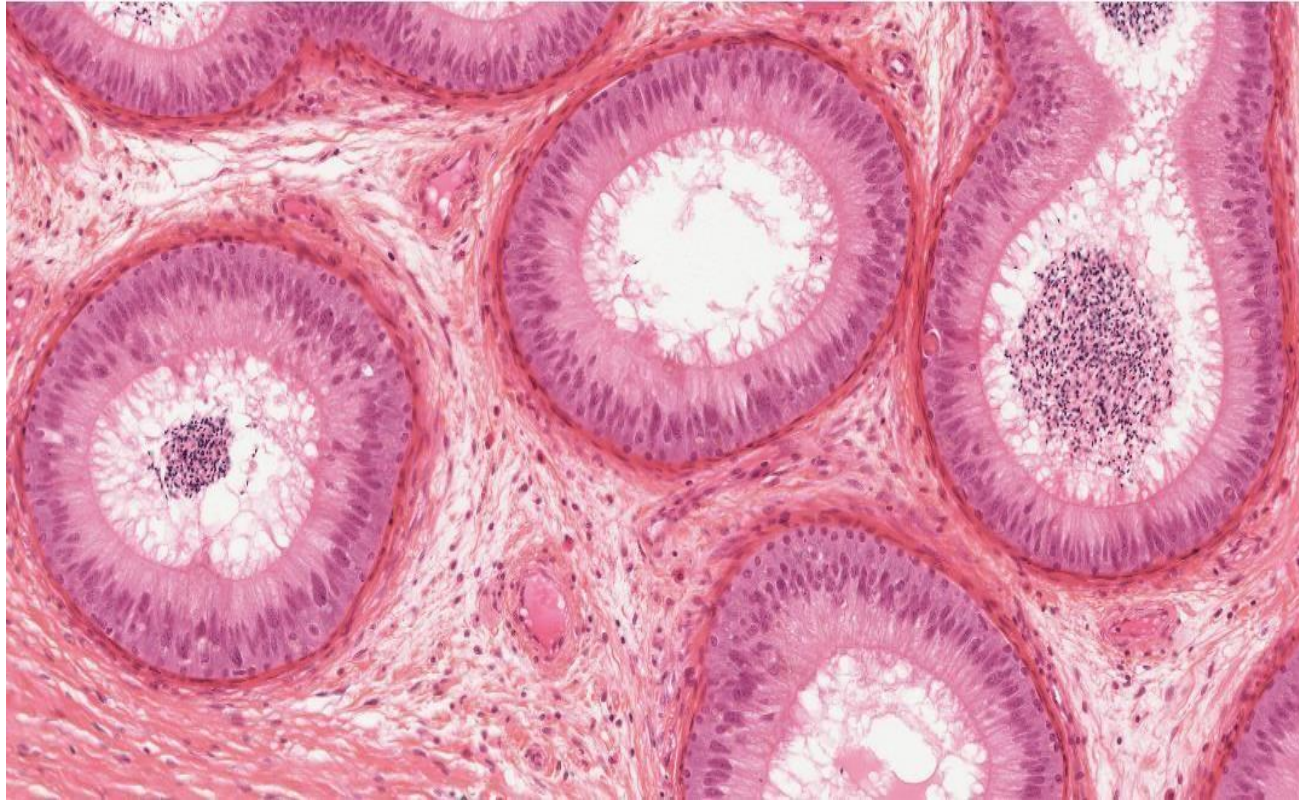


Efferent ductules

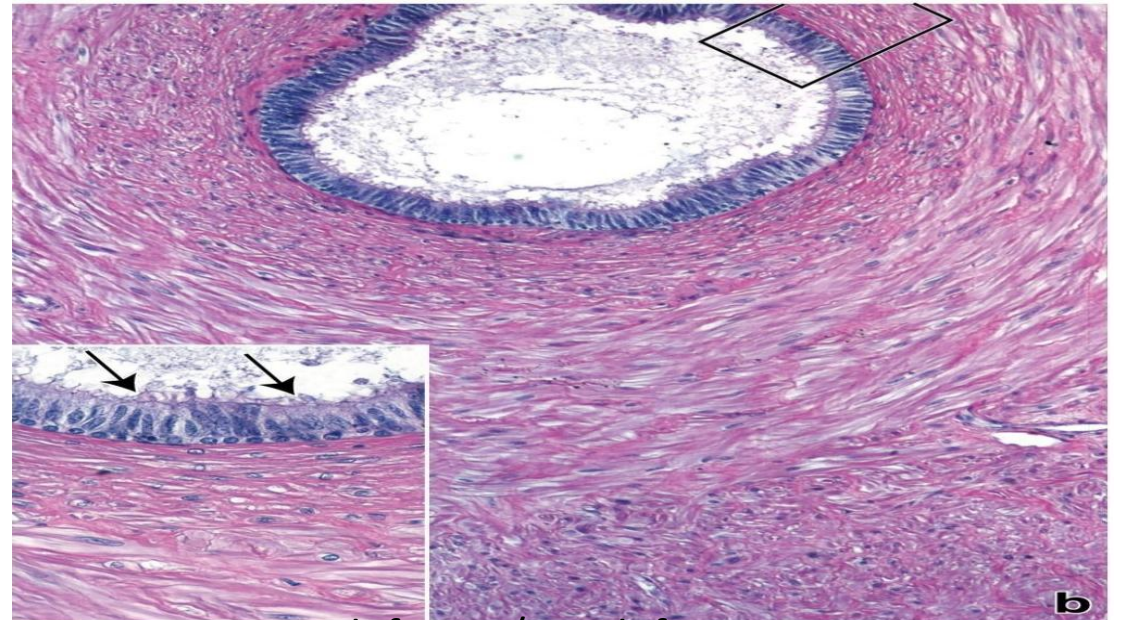
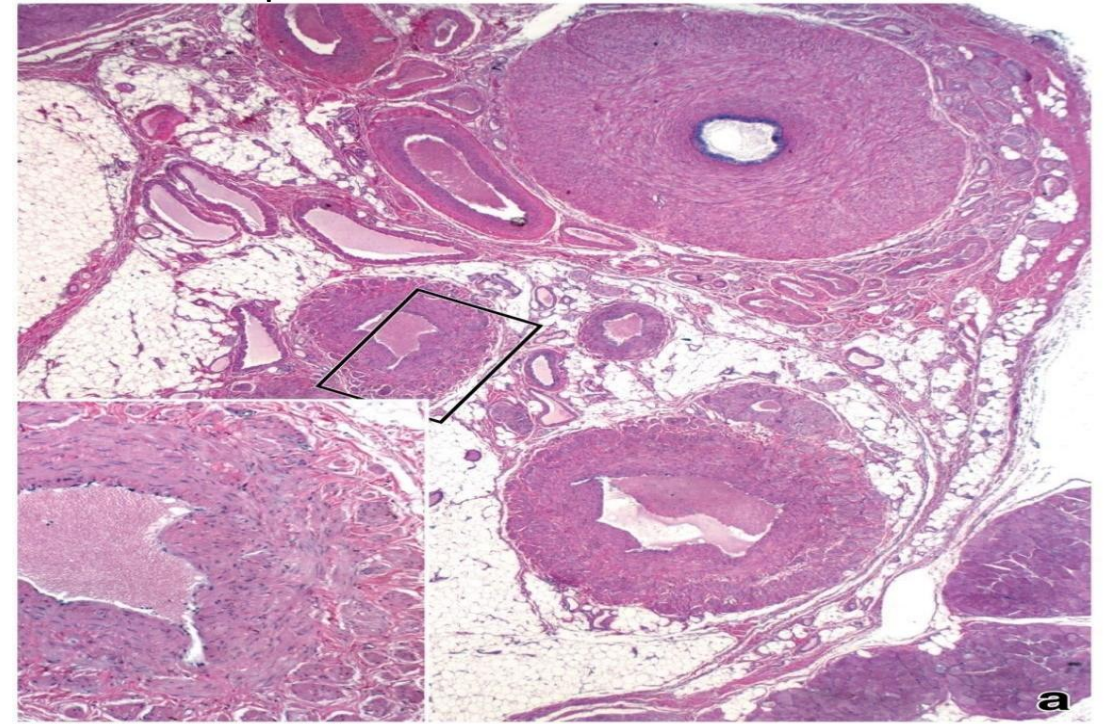




Epididymis



Spermatic Cord



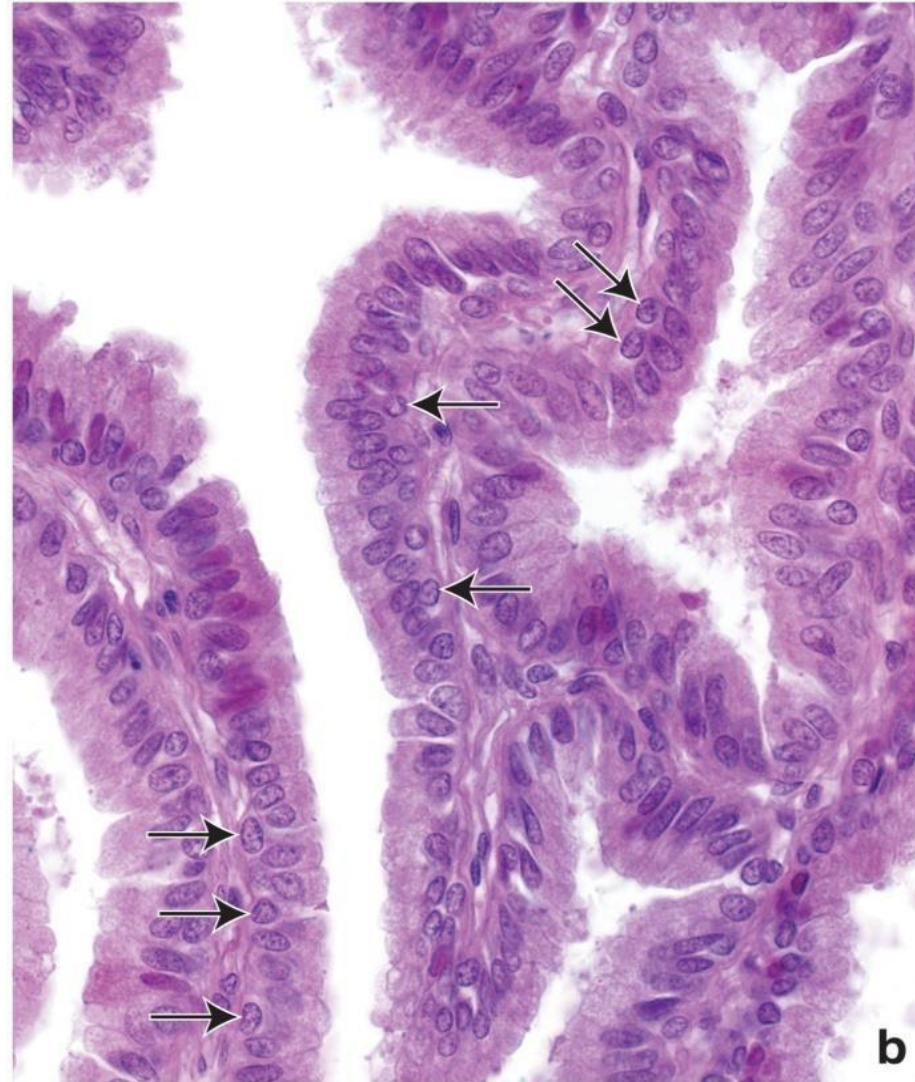
Ductus deferens/Vas deferens



## Seminal Vesicle



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# Questions: Integrated Case 1 Continued

4. Review the histology of the path taken by sperm from production to ejaculation.
5. What is the blood supply, venous drainage and lymphatic drainage of the testes?
6. What is the innervation of the testes?
7. (a) What would you expect to find on the scrotal ultrasound?  
(b) Explain how this/these finding(s) would lead to infertility.



# Integrated Case 2

A 56-year-old man comes to the physician because of a 5-month history of loss of appetite, weight loss, and blood in the urine (hematuria). As of recently he has also noticed difficulty with passing urine. A digital rectal examination shows an enlarged, nodular and painless prostate. Urinalysis shows macroscopic hematuria.

1. What is the relationship of the prostate to the bladder, rectum, pelvic diaphragm, perineal membrane, urethra, seminal vesicle and vas deferens?
2. Review the histology of the prostate gland.





# Questions: Integrated Case 2 Continued

3. Which region of the prostate is most likely diseased to result in the symptoms described?
4. Discuss the blood supply of the prostate gland.
5. What other conditions can produce an elevated PSA?

**The patient undergoes an open prostatectomy to remove the tumor. At his 3-month follow up examination, he complains of not being able to receive an erection.**

6. Discuss the processes of erection and ejaculation and explain why his erectile dysfunction can be a post-surgical complication.